MPCA Project

TITLE OF THE PROJECT:

Team Members:

Sl. No	Name	SRN
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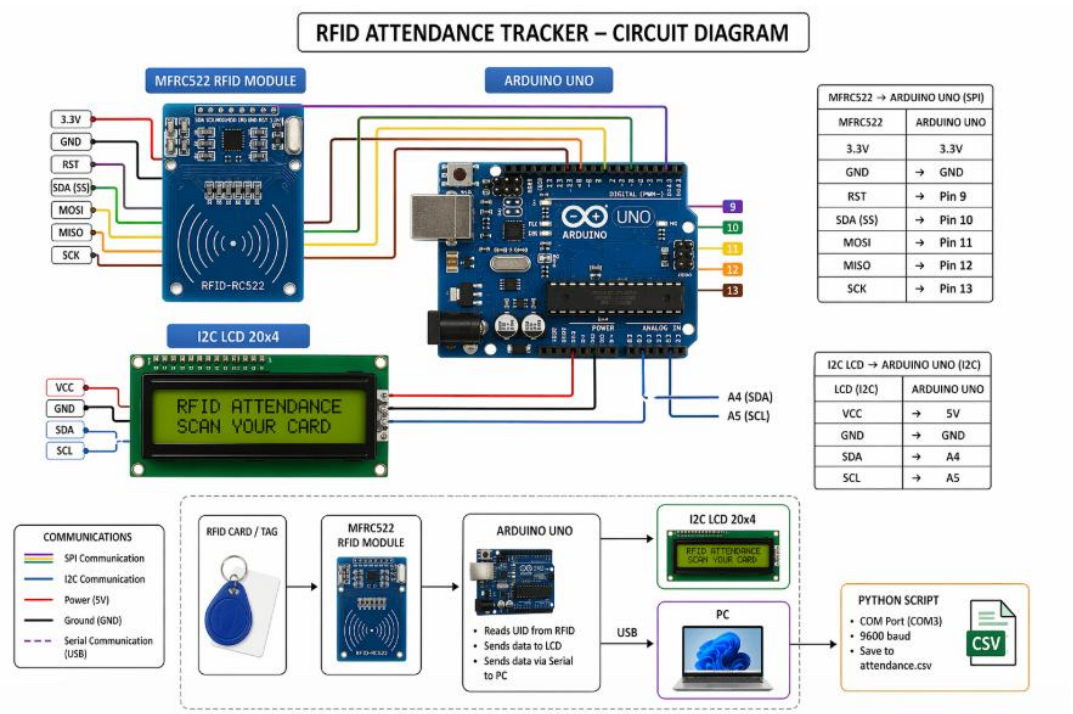
Components used:

Component	Specification	Purpose
Arduino UNO	ATmega328P microcontroller	Main controller
MFRC522 RFID Module	13.56 MHz, SPI interface	Card reading
RFID Cards/Tags	ISO 14443A, unique UID	Student identification
I2C LCD Display	20×4 characters, 0x27 address	Real-time feedback
Jumper Wires	Male-to-male/female	Circuit connections
USB Cable	Type A to Type B	Arduino-PC serial link

Explanation about the project:

This project presents an **RFID-based attendance tracking system** that integrates Arduino UNO hardware with a Python-based software layer. Each student is assigned a unique RFID card. When a card is scanned, the Arduino reads the card's UID, displays the student's name and attendance status on a LCD display, and transmits the UID via serial communication to a connected computer. The Python script receives this UID, logs the student's name, time of attendance, and assignment submission status into a CSV file. The system prevents duplicate entries within a session and generates a final attendance summary at the end. This project demonstrates the integration of embedded hardware and software to build a real-world automated attendance solution.

Circuit Diagram:



Pseudocode:

1)Arduino

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define SS_PIN 10
#define RST_PIN 9
MFRC522 mfrc522(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 20, 4);

String student1 = "3C E9 FA 03";
String student2 = "D5 CD FE 03";
String student3 = "91 99 FB 03";
String student4 = "CC EF FF 03";
bool marked1 = false;
bool marked2 = false;
bool marked3 = false;
bool marked4 = false;
void setup() {
  Serial.begin(9600);
  SPI.begin();
  mfrc522.PCD_Init();
  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("RFID Attendance");
  lcd.setCursor(0, 1);
  lcd.print("Scan your card");
}
void loop() {
  if (!mfrc522.PICC_IsNewCardPresent()) return;
  if (!mfrc522.PICC_ReadCardSerial()) return;
  String content = "";

  for (byte i = 0; i < mfrc522.uid.size; i++) {
    content += String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
    content += String(mfrc522.uid.uidByte[i], HEX);
```

```
  }
  content.toUpperCase();
  Serial.println(content.substring(1));
  lcd.clear();
  if (content.substring(1) == student1) {
    lcd.setCursor(0, 0);
    lcd.print("Aryan");
    lcd.setCursor(0, 1);
    lcd.print(marked1 ? "Already Marked" : "Attendance Done");
    marked1 = true;
  }
  else if (content.substring(1) == student2) {
    lcd.setCursor(0, 0);
    lcd.print("Shetty");
    lcd.setCursor(0, 1);
    lcd.print(marked2 ? "Already Marked" : "Attendance Done");
    marked2 = true;
  }
  else if (content.substring(1) == student3) {
    lcd.setCursor(0, 0);
    lcd.print("Vinju");
    lcd.setCursor(0, 1);
    lcd.print(marked3 ? "Already Marked" : "Attendance Done");
    marked3 = true;
  }
  else if (content.substring(1) == student4) {
    lcd.setCursor(0, 0);
    lcd.print("Aditya");
    lcd.setCursor(0, 1);
    lcd.print(marked4 ? "Already Marked" : "Attendance Done");
    marked4 = true;
  }
  else {
    lcd.setCursor(0, 0);
    lcd.print("Unknown Card");
  }
  delay(2000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Scan your card");
}
```

2) Python Code

```
import serial
import csv
from datetime import datetime
import os
import time

PORT = 'COM3'
BAUD = 9600
students = {
    "3C E9 FA 03": {"name": "Aryan", "assignment": "Submitted"},
    "D5 CD FE 03": {"name": "Shetty", "assignment": "Pending"},
    "91 99 FB 03": {"name": "Vinju", "assignment": "Submitted"},
    "CC EF FF 03": {"name": "Aditya", "assignment": "Pending"}
}
marked = set()
filename = "attendance.csv"
if not os.path.exists(filename):
    with open(filename, "w", newline='') as file:
        writer = csv.writer(file)
        writer.writerow(["Name", "UID", "Time", "Assignment Status"])
ser = serial.Serial(PORT, BAUD)
time.sleep(2)
print("System Ready... Scan cards\\n")

try:
    while True:
        if ser.in_waiting > 0:
            uid = ser.readline().decode(errors='ignore').strip()
            if not uid:
                continue
            print("Scanned UID:", uid)

            if uid in students:
                name = students[uid]["name"]
                assignment = students[uid]["assignment"]
                if uid not in marked:
                    marked.add(uid)
                    time_now = datetime.now().strftime("%H:%M:%S")
                    with open(filename, "a", newline='') as file:
                        writer = csv.writer(file)
                        writer.writerow([name, uid, time_now, assignment])
```

```
                print(name, "marked PRESENT at", time_now)
                print("Assignment Status:", assignment, "\\n")
            else:
                print(name, "already marked\\n")
        else:
            print("Unknown card\\n")
except KeyboardInterrupt:
    print("\\nSESSION ENDED\\n")
    print("Attendance Summary:")
    for uid, data in students.items():
        status = "Present" if uid in marked else "Absent"
        print(data['name'], "->", status)
    print("\\nFile saved as:", filename)
```

Output Pictures with explanations:

Final output in the csv file:

	A	B	C	D	E
1	Name	UID	Time	Assignment Status	
2	Aditya	CC EF FF 0:	14:16:20	Pending	
3	Shetty	D5 CD FE C	14:16:28	Pending	
4	Vinju	91 99 FB 0	14:16:30	Submitted	
5	Aryan	3C E9 FA 0	14:16:33	Submitted	
6	Aryan	3C E9 FA 0	14:20:26	Submitted	
7	Aditya	CC EF FF 0:	14:21:29	Pending	
8	Shetty	D5 CD FE C	14:21:31	Pending	
9	Vinju	91 99 FB 0	14:21:41	Submitted	
10					